Lilian **Hunout**

Machine Learning Engineer

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Work Experience

Criteo — **Paris, France**

Machine Learning Engineer II — Oct. 2025 - Current

- Reco Models team
- Designed and managed end-to-end ML pipelines for recommendation sources (retargeting and contextual), including data preparation with lineage tracking, model training, and automated deployment
- Implemented state-of-the-art models with novel feature engineering and user modeling approaches for improved retrieval performance
- Configured A/B tests at scale (up to worldwide traffic), conducted statistical analysis, and managed production server pools for high-traffic environments
- Acted as on-call engineer to troubleshoot critical issues in offline and online production systems
- Mentored engineers and created comprehensive documentation to foster team knowledge sharing

Machine Learning Engineer — Oct. 2024 - Sep. 2025

Machine Learning Research Intern (AI Lab) — Apr. 2024 - Sep. 2024

- Reco Research team, supervised by David Rohde
- Subject: Optimizing Onsite Display Advertising through Deep Learning

Eviden — **Grenoble, France**

AI & Computer Vision Research Intern (Atos BDS - CV Lab) — Feb. 2023 - Aug. 2023

- European project PREVENT PCP
- Detection of abandoned luggage in video streams to improve station safety and limit false alerts
- Implementation of score-based object/owner association algorithms, object tracking and re-identification

MBDA — **Bourges, France**

Embedded AI Intern (Open Lab) — Apr. 2022 - Sep. 2022

- Implementation of communication protocols, creation of an embedded Linux image on a Xilinx Zynq card, setting up a Vitis AI Docker environment, model benchmarking

Education

Ecole Normale Supérieure Paris-Saclay — **Paris, France**

MSc in Mathematics, Vision, Apprentissage (MVA) — Sep. 2023 - Sep. 2024

- World's leading #1 university in Mathematics (ARWU)
- Courses: Deep Learning (theory and practice), Computer Vision, Object Recognition, Reinforcement Learning, Time Series Analysis, Natural Language Processing, Speech Processing, Convex Optimization, Probabilistic Graphical Models, Deep Generative Models, Signal Processing
- Representative of a class of 250+ students

INSA Centre Val de Loire — **Bourges, France**

MEng in Industrial Systems — Sep. 2018 - Sep. 2023

- Courses: Statistical Theory and Methods, Machine Learning, Computer Vision, Complex Systems Engineering, Risk Management
- Student representative, active participation in various events (student council campaign, open days, etc.)

University of Tours — **Tours, France**

MSc in Applied Physics (Double Degree) — Sep. 2022 - Sep. 2023

- Courses: Automatics, Robotics, Cyber-Physical Systems, Multiphysics Modelling

Tallinn University of Technology — **Tallinn, Estonia**

School of Information Technologies (Erasmus) — Sep. 2021 - Jan. 2022

- Courses: Data Mining, Machine Learning, System Programming, Embedded Systems

Skills

Programming	Python, SQL, Scala, C, MATLAB/Simulink
ML Frameworks	PyTorch, PyTorch Lightning, Hugging Face Transformers, TensorFlow, Scikit-learn, OpenCV, Gradio
Big Data & Distributed Computing	PySpark, Ray, Apache Hadoop, CUDA, Docker, SLURM
Data Processing	Pandas, NumPy, MLflow, Hydra, Matplotlib
Tools & Others	Git, Jupyter, LaTeX/Markdown, Jira/Confluence, Microsoft Office
Soft Skills	Problem-solving, teamwork, communication, time management, documentation
Languages	English (professionnal), French (native)

Projects

ENS Paris-Saclay **Paris, France**

MSc Projects Fall 2023 - Winter 2024

- Challenge Data Qube Research & Technologies (Collège de France): implementation of a football match results prediction model
- Implementation of a contrastive learning technique to enable the retrieval of molecules from natural language queries
- Automating video segmentation with ”Segment Anything Meets Point Tracking”
- Partial reimplementation of the paper ”Deep Temporal Clustering: Fully Unsupervised Learning of Time-Domain Features”, analysis and testing

INSA Centre Val de Loire **Bourges, France**

PRISME Laboratory Spring 2022

- Detection and tracking of micro-scale magnetic robots on an ultrasound dataset (medical applications)

TalTech Embedded AI Research Lab **Tallinn, Estonia**

ML for Embedded Systems Course Winter 2022

- Implementation of a garage door cycle detection model from an accelerometric data set on an STM32 board with X-CUBE-AI. Award: first prize 250€